

[Core] 6.1 Example t_2 and $U_{4,2}$.

We have FD Complexes $\mathfrak{T}_2$ for t_2 and $\mathfrak{U}_{4,2}$ for $U_{4,2}$.

$$E(\mathfrak{T}_2) = E(\mathfrak{U}_{4,2}) = \{1,2,3,4\}$$

$$\mathcal{I}(\mathfrak{T}_2) = \mathcal{I}(\mathfrak{U}_{4,2}) = \mathcal{P}_{\leq 2}(\{1,2,3,4\})$$

$$\max \mathfrak{T}_2 = \{\{\emptyset\} \rightarrow \{\emptyset\}, \{1\} \rightarrow \{1\}, \{2\} \rightarrow \{2\}, \{3\} \rightarrow \{3\}, \{4\} \rightarrow \{4\}, \{1,2\} \rightarrow \{E\}, \{1,3\} \rightarrow \{E\}, \{1,4\} \rightarrow \{E\}, \{2,3\} \rightarrow \{E\}, \{2,4\} \rightarrow \{E\}, \{3,4\} \rightarrow \{3,4\}\}$$

$$\subseteq (\{\{\emptyset\} \rightarrow \{\emptyset\}, \{1\} \rightarrow \{1\}, \{2\} \rightarrow \{2\}, \{3\} \rightarrow \{3\}, \{4\} \rightarrow \{4\}, \{1,2\} \rightarrow \{E\}, \{1,3\} \rightarrow \{E\}, \{1,4\} \rightarrow \{E\}, \{2,3\} \rightarrow \{E\}, \{2,4\} \rightarrow \{E\}, \{3,4\} \rightarrow \{E\}\})^+$$

$$= (\max \mathfrak{U}_{4,2})^+ = F(\mathfrak{U}_{4,2})$$

by Corollary **[KyCl] 4.10**.

□

We are ready to prove the other definitive properties of Core FD Complexes. The tetrahedral FD Complex $\mathfrak{T}_2$ does not satisfy (FD 4) and already demonstrate failure of these properties.

[Core] 6.2 Theorem: Core FD Complexes are Well Behaved

The following are equivalent:

1. $\mathfrak{F}$ is a Core FD Complex.
2. If $K \in \mathcal{I}_{\mathfrak{F}}$ is an independent subset of $X \subseteq E$ then there is a key $K_X \in \mathcal{I}_{\mathfrak{F}}$ of X with $K \subseteq K_X$.
3. If I is a maximal independent subset of $X \subseteq E$ then I is a key of X .
4. Every md-set in E is circuit.
5. If C is an md-set, $e \in C$ and $C - e \subseteq S$ a closed then $e \in S$.

Proof:

(1) $\Rightarrow$ (2): Let $n_{K,X} = |X - \phi_F(K)|$. We will induct on $n_{K,X}$. If $n_{K,X} = 0$ then $K = K_X$.

Suppose there is a desired K_X whenever $0 \leq n_{K,X} < n$, and suppose $n_{K,X} = n$. There is an $e \in X - \phi_{\mathfrak{F}}(K)$. $e \notin \phi_{\mathfrak{F}}(K)$ implies $K \rightarrow \{e\} \notin F_{\mathfrak{F}}$ by **Lemma [KyCl] 4.5(c)**. We have, $K \in \mathcal{I}_{\mathfrak{F}}$, $e \in E_{\mathfrak{F}} - K$ and $K \rightarrow \{e\} \notin F_{\mathfrak{F}}$. By the strong extension (FD 4 contra.) $K \cup e \in \mathcal{I}_{\mathfrak{F}}$. This means that $K \cup e$ is an independent subset of $X \cup e$,